Aram Davtyan

Bern, Switzerland
araachie@gmail.com
araachie.github.io
araachie

Technical Expertise

- Generative AI
- Video/Image Generation
- Self/Unsupervised Learning
- World Models
- Controllable Generation
- Representation Learning

Education

Sep 2020 – Dec 2024 **Ph.D. in Computer Science (summa cum laude)**,
University of Bern, *Computer Vision Group*, Bern, Switzerland.
Thesis: Learning Object Interactions via Efficient and Controllable Generative Models.
Supervisor: Prof. Dr. Paolo Favaro

Sep 2014 – Jun 2020 **B.S. + M.S. equivalent in Mathematics (with honors, GPA 4.98/5)**,
Lomonosov Moscow State University, Moscow, Russia.
Thesis: Spectral analysis of the operator in the generalized 't Hooft model.
Supervisor: Prof. Dr. Igor Sheipak

Sep 2016 – Jun 2018 **M.S. equivalent in Data Analysis**,
Yandex School of Data Analysis, Moscow, Russia.
Thesis: Optimization of oil production via forecast models on injection wells operation modes.
Supervisor: Prof. Dr. Ilya Muchnik

Job Experience

Jan 2025 – present **Postdoctoral Researcher**,
University of Bern, *Computer Vision Group*, Bern, Switzerland.
- Research in Generative AI, World Models, and (Controllable) Video Generation.
- Conducting large-scale generative model training as part of the Swiss AI Initiative.

Jun 2024 – Sep 2024 **Research Intern**,
Disney Research Studios, Zürich, Switzerland.
- Developed DL models for video quality enhancement (super-resolution, deinterlacing, stabilization).

Feb 2019 – Sep 2020 **Software Engineer**,
Yandex, Moscow, Russia.
- Developed backend for Yandex.Navigator, a high-load routing service using real-time traffic.
- Built data acquisition pipelines, improved data quality and routing algorithms on large road graphs.
- Significantly improved suggested route quality via supervised and reinforcement learning.

Sep 2017 – Dec 2021 **ML Specialist**,
Inline group, Moscow, Russia.
- Developed ML solutions for the oil industry: forecasting production rates and analyzing well relationships.
- Led data preprocessing, customer communication, and solution presentation.

Skills

Programming	Python, C/C++
ML/DL	PyTorch, torchvision, Transformers, Diffusion, Generative Models, TensorFlow, NumPy, pandas, matplotlib, sklearn
Tools	Git, SQL, MapReduce, LaTeX
Mathematics	Functional Analysis, Probability Theory, Statistics

Awards and Honors

Dec 2025 **ARC Prize - Honorable Mention**, *ARC Prize Foundation*.
For research demonstrating that Video Diffusion Models (VDMs) can be repurposed to solve a wide range of novel visual and logic tasks with only handful of examples.

Talks and Presentations

Sep 2025 **Swiss AI Initiative: Our Experience at Computer Vision Group @ UniBE** , *Bern, Switzerland*.
"AI for SMEs: What can small buisnesses really do with artificial intelligence?" Workshop, Swiss AI Weeks.

Sep 2024 **Unsupervised Controllable Video Generation**, *Zürich, Switzerland*.
Invited Seminar Talk, Computer Vision Group, ETH Zürich.

Sep 2023 **Efficient Video Prediction via Sparsely Conditioned Flow Matching**, *Heidelberg, Germany*.
Nectar Track Oral Presentation, GCPR 2023.

Teaching

Spring 2025 **Lecturer**, *University of Bern*, Bern, Switzerland.
"Seminar Machine Learning and Artificial Intelligence" (M.S. level)

Fall 2025 **Lecturer**, *University of Bern*, Bern, Switzerland.
"Foundations of Deep Learning" (M.S. level) course taught jointly with Prof. Dr. Andrea Agazzi

Falls 2020-2024 **Teaching Assistant**, *University of Bern*, Bern, Switzerland.
"Machine Learning" (B.S. level) course taught by Prof. Dr. Paolo Favaro

Springs 2023-2025 **Teaching Assistant**, *University of Bern*, Bern, Switzerland.
"Deep Learning" (M.S. level) course taught by Prof. Dr. Paolo Favaro

Fall 2021 **Teaching Assistant**, *Yandex School of Data Analysis*, Moscow, Russia.
Seminar on Self-Supervised learning in Computer Vision for the "Machine Learning" course.

Spring 2021 **Teaching Assistant**, *University of Bern*, Bern, Switzerland.
"Advanced Topics in Machine Learning" (M.S. level) course taught by Prof. Dr. Paolo Favaro

Supervision

Sep 2025 **Nicola Irmiger**, *Towards Multimodal Apertus*, Master Project, ETH Zürich.

Sep 2024 **Darya Ardan**, *Analysis of the Curvature of the Trajectories in the Concept of Flow Matching for Generative Models*, Master Project, University of Bern.

Sep 2023 **Adrian Schmucker**, *Generating Layered Image Representations with Diffusion Models Using Score Distillation Sampling*, Master Project, University of Bern.

Jun 2020 **Alexander Rodin**, *Forecast Models of Oil Production Rates Based on Sliding Window Regression*, Diploma, Yandex School of Data Analysis.

Jun 2020 **Alexey Kuchaev**, *Application of Machine Learning Methods for Evaluating Hydrodynamic Relationships between Wells*, Diploma, Yandex School of Data Analysis.

Jun 2020 **Aleksandr Khalaidzhi**, *Construction of an Oil Production Forecast Model via Incorporating Interwell Relationships*, Diploma, Yandex School of Data Analysis.

List of Publications

Preprints

Mar 2026 **Communication-Inspired Tokenization for Structured Image Representations**, *arXiv*.
Aram Davtyan, Yusuf Sahin, Yasaman Haghighi, Sebastian Stapf, Pablo Acuaviva, Alexandre Alahi, Paolo Favaro.
[paper] [website] [github]

Jun 2025 **From Generation to Generalization: Emergent Few-Shot Learning in Video Diffusion Models**, *arXiv*.
Pablo Acuaviva, ***Aram Davtyan***, Mariam Hassan, Sebastian Stapf, Ahmad Rahimi, Alexandre Alahi, Paolo Favaro.
[paper] [website] [github]

Conference Proceedings

Jul 2026 **Rethinking Visual Intelligence: Insights from Video Pretraining**, *ICML 2026*.
Pablo Acuaviva, ***Aram Davtyan***, Mariam Hassan, Sebastian Stapf, Ahmad Rahimi, Alexandre Alahi, Paolo Favaro.
[paper]

Apr 2026 **Composition of Memory Experts for Diffusion World Models**, *ICLR 2026*.
Sebastian Stapf, Pablo Acuaviva, ***Aram Davtyan***, Paolo Favaro.
[website]

Feb 2026 **Learning Vision–Language Alignment in Unified LLMs with 24 Text Tokens per Image**, *IWSDS 2026*.
Nicola Irmiger, Yixuan Xu, Raphael Kreft, ***Aram Davtyan***, Manuel Kaufmann, Imanol Schlag.

Dec 2025 **KOALA++: Efficient Kalman-Based Optimization of Neural Networks with Gradient-Covariance Products**, *NeurIPS 2025*.
Zixuan Xia, ***Aram Davtyan***, Paolo Favaro.
[paper]

Oct 2025 **MIRAGE: Unsupervised Single Image to Novel View Generation with Cross Attention Guidance**, *ICCVW 2025*.
Llukman Cerkezi, ***Aram Davtyan***, Sepehr Sameni, Paolo Favaro.
[paper] [github]

Jun 2025 **GEM: A Generalizable Ego-Vision Multimodal World Model for Fine-Grained Ego-Motion, Object Dynamics, and Scene Composition Control**, *CVPR 2025*.
Mariam Hassan, Sebastian Stapf, Ahmad Rahimi, Pedro M B Rezende, Yasaman Haghighi, David Brüggemann, Isinsu Katircioglu, Lin Zhang, Xiaoran Chen, Suman Saha, Marco Cannici, Elie Aljalbout, Botao Ye, Xi Wang, ***Aram Davtyan***, Mathieu Salzmann, Davide Scaramuzza, Marc Pollefeys, Paolo Favaro, Alexandre Alahi.
[paper] [website] [github]

Apr 2025 **Faster Inference of Flow-Based Generative Models via Improved Data-Noise Coupling**, *ICLR 2025*.
Aram Davtyan, Leello Tadesse Dadi, Volkan Cevher, Paolo Favaro.
[paper] [website]

Feb 2025 **CAGE: Unsupervised Visual Composition and Animation for Controllable Video Generation**, *AAAI 2025*.
Aram Davtyan, Sepehr Sameni, Björn Ommer, Paolo Favaro.
[paper] [website] [github]

Feb 2024 **Learn the Force We Can: Enabling Sparse Motion Control in Multi-Object Video Generation**, *AAAI 2024*.
Aram Davtyan and Paolo Favaro.
[paper] [website] [github]

Oct 2023 **Efficient Video Prediction via Sparsely Conditioned Flow Matching**, *ICCV 2023*.
Aram Davtyan, Sepehr Sameni, Paolo Favaro.
[paper] [website] [github]

Oct 2022 **Controllable Video Generation through Global and Local Motion Dynamics**, *ECCV 2022*.
Aram Davtyan and Paolo Favaro.
[paper] [website] [github]

Feb 2022 **KOALA: A Kalman Optimization Algorithm with Loss Adaptivity**, *AAAI 2022*.
Aram Davtyan, Sepehr Sameni, Llukman Cerkezi, Givi Meishvili, Adam Bielski, Paolo Favaro.
[paper] [website] [github]

Journals

Dec 2020 **Oil Production Forecast Models Based on Sliding Window Regression**, *Journal of Petroleum Science and Engineering 2020*.
Aram Davtyan, Alexander Rodin, Ilya Muchnik, Alexey Romashkin.
[paper]

Patents

Oct 2019 **Control method of oil production at mature separate oil deposit**, *RU2701761C1*.
Aram Davtyan, Alexey Romashkin, Ilya Muchnik.
[url]